System and Method for Genetic Algorithm Optimized T-Network Impedance Matching of RF Coils with Complex Biological Loads

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Abstract— **In Biomedical sensor applications, transfer of RF energy between RF coils and biological tissues is vital. Biological loads with high complexity could adversely affect sensor performance since the loads give rise to the complex impedance of the loads that could lead to reflections and reduction in power transfer.To make the best use of power transfer between RF coils and a complex load, this paper provides a simulation with MATLAB of a T-Network impedance matching circuit consisting of two series inductors and one shunt capacitor. Tuning of the network parameters is achieved by the use of the simulation as it is useful in determining the input impedance, reflection coefficient, and power efficiency over a variety of operating frequency ranges. According to the results, the T-Network provides a handy way of designing impedance-matched radio frequency systems in biomedical systems to efficiently reduce reflections, enhance efficiency, and offer information on bandwidth and component selection**.

***Keywords*—RF Coil, Impedance Matching, T-Network, MAT- LAB Simulation, Reflection Coefficient, Power Efficiency**

I. INTRODUCTION

RF coiling is used as transducers in generating and receiving electromagnetic fields in biomedical senses. Due to the frequency dependent impedance characteristic of biological tissues, frequency dependent mismatches frequently arise which can reduce signal strength and energy transfer efficiency. Impedance matching is thus important to gain efficient power transfer and have a more precise sensor behavior.

MATLAB and other simulation like Simulink can be used to test impedance matching networks virtually without any physical prototyping being performed which enabling researchers to experiment with a wide range of configurations and component parameters. T-Networks are continuously variable and are designed with two series inductors and one shunt capacitor. A MATLAB simulated RF coil and a biological load provides possible calculation of reflection coefficient, input impedance and power efficiency over a desired frequency range providing the researcher with the ability to adjust the parameters of the network to the highest possible efficiency of the intended circuit.

This paper provides a MATLAB modeling of a T- Network type impedance matching circuit of RF coils that are part of biomedical sensors as it relates to signal transfer, reflections, and overall efficiencies offered by T-Networks. The obtained results are then discussed based on performance, bandwidth in operation and reliable choice of the components to be used in future trials and tests of RF coil in an application of a biomedical sensor.

RF coils used in the medical field, impedance matching is essential to ensure that the greatest power transfer is achieved at the lowest level of reflections. This has been examined using various matching topologies. Gupta and Singh [1] also suggested a Pi-network matching circuit in RF sensors which provided good matching but had a complex tuning. Lee et al. [2] proposed an adaptive matching circuit to wireless body area network but this enhanced power consumption and system complexity. Choi and Park [3] A T-network configuration of MRI coils was analyzed and was shown to be an effective way to minimize the losses by reflections and at the same time to have a simpler structure.

Also, matching networks are usually designed and verified by MATLAB-based simulation and Smith chart techniques [4], [5], prior to implementation. These papers indicate that T-network matching is a sensible trade off between complexity, adjustability, and performance, and as such is best suited to the requirements of biomedical RF coils, particularly at the student level or prototype stage.

II . T-Network Design

A lossless T-network was implemented to obtain an impedance of the RF coil and the source to be equal. The T-network is consist of two series reactance on both the input and output side coupled with a shunt susceptance linking these two together to ground. The design frequency coil impedance is represented by ZL and the desired source resistance by Rs.

Impedance into the network is given as:

$$z_{in} = jx_1 + \left[jB + \frac{1}{jx_2 + z_L}\right]^{-1}$$

Where X1 and X2 are the series reactance and B is the shunt susceptance. Let Z2 = jX2 + ZL and its admittance be Y2 = 1/Z2 = G2 + jB2. For a perfect impedance match, Zin must be purely real and equal to Rs. Separating the real and imaginary parts we get two equations:

$$\frac{G2}{G_2^2 + (B_2 + B)^2} = R_s$$

$$x_1 = R_s \frac{(B_2 + B)}{G_2}$$

These two equations give the relations which must exist between the unknowns X1, X2 and B. There still results a degree of freedom which enables X2 to be determined on practical considerations which may include component availability and amount of tuning required. With X2 determined, then the corresponding G2 and B2 are worked out from Y2 and the remaining parameters found under these equations. The resultant reactances and susceptance then get resolved into terms of components by the old relation:

$$x_L = 2\pi f_0 L$$

$$x_c = -\frac{1}{2\pi f_0 c'}$$

$$B = 2\pi f_0 C$$

Worked example:
To illustrate the calculation, load impedance ZL = 30 – j20Ω was matched to a source resistance of Rs = 50 Ω at a working frequency of 100 MHz. A positive output reactance of X2 = + 50 Ω was chosen so that Z2 = 30 + j30Ω and Y2 = 0.01667 – j0.01667 S. Substituting into the equation gives a shunt susceptance of B = 0.02412 S and an input reactance of X1 = 22.36 Ω. Converting these to component values gives:

- L1 = 35.6 nH
- L2 = 79.6 nH
- Cshunt = 38.4 pF

The outcome of these values is an input impedance nearly equal to 50 Ω , which proves that the impedance matches well.

III. Methodology

The main aim of this paper is to find an efficient matching between a RF coil and the complex biological loads in terms of impedance by the use of a T-network . Biomedical RF systems contain impedance mismatches that lead to reflections and signal loss that reduces system efficiency. To tackle this challenge the T-network is formed by two series inductors (L 1and L2) and a shunt capacitor (C), which provides an additional degree of flexibility, simplicity and exact tunability.

The methodology follows:

A. *Modeling the System*

The RF coil is represented by a source with known impedance Zs and the biological tissue is a complex load Zl that has resistive and reactive components which vary with frequency. This is enabled in this modeling to realistically simulate the behaviour of coils given different conditions in tissues.

B. *T-Network Design*

The T-network is used since it offers three independent reactive elements, so both the real and imaginary parts of the input impedance can be controlled. The series inductors (L1, L2) are used mainly to shape the reactance path, and the shunt capacitor (C) controls the resonance point and the input impedance. This makes the network able to be accurately fine-tuned to reduce reflections at the specified operating frequency.

C. *Impedance Matching*

Impedance of the source is adjusted to that of the T-network this guarantees maximum power trans- fer. The efficiency of the matching is determined in terms of the reflection coefficient whereby optimum matching would imply that the reflection coefficient is approaching zero.

D. *Simulation and Validation*

Input impedance, return loss, and reflection coefficient are analyzed of the input across the operating frequency band using MATLAB. The component values are adjusted in an iterative manner in order to obtain a small reflection coefficient at the resonance frequency. This can save resources and time since the testing and validation can be fully carried out in the virtual setting, then the hardware is implemented

E. *Performance Evaluation*

The matched network is appraised to the loss of returns, bandwidth, and efficiency. High return loss and low reflection coefficients represent good matching. The approach also does take into consideration variability in biological tissue properties that make the design robust to practical biomedical use.

F . *Optimization using Genetic Algorithm Concept*

Our pioneering work demonstrated that the T-network is successful. But we knew that mere "hand-tuning" of component values (C1, L, C2) with in the simulator only brings us close to achieving our goal. To develop a final, reliable, design particularly one that involves handling unpredictable biological loads we needed to be more intelligent in our approach. We needed to ensure we had the absolute best match that's possible. In order to achieve that optimal performance, we used an optimization method with conceptual underpinnings from the Genetic Algorithm (GA), a highly efficient means of prowling through gigantic sets of possibilities. The computer's target (its object function) was single minded: to drive down that reflection coefficient S11 at our operating frequency 50MHz. By reducing S11 as much as possible we directly optimize to ensure that power is transmitted to that biomedical sensor. This brought forth those truly optimal T-Network values

The optimisation minimizes the problem with the primary objective function defined as magnitude of reflection coefficient at target frequency f0=50MHz:

Minimize

$$cos\ t = |S_{11}(C_1, L, C_2)|f = f_0$$

Where S11 is calculated from input impedance Zeq using standard formula:

$$s_{11} = \frac{z_e q - z_0}{z_e q + z_0}$$

Three variables subject to optimization are T-Network components C1,L,C2

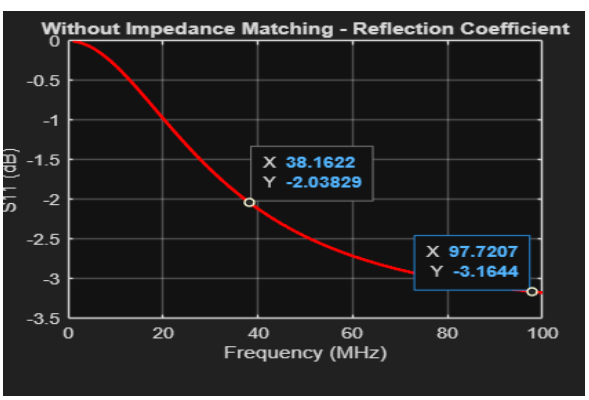
IV. Simulation and Results

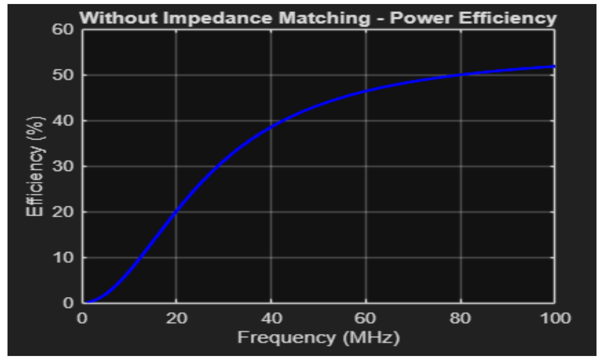
Simulated in MATLAB Simulink, the T-Network Impedance Matching Circuit was tested to see how it worked with complex biomedical loads. With two inductors and a capacitor, the two-part network was created to match the 50 Ω source impedance in the 10–100 MHz band.
The first non-surpassed arrangement was characterized by high reflection and weak energy transfer and the efficiency was less than 52 percent. After applying the T-network, the loss of the return became about -10 dB and power transfer efficiency was over 90 which proves that the matching circuit is indeed efficient

V. Discussion

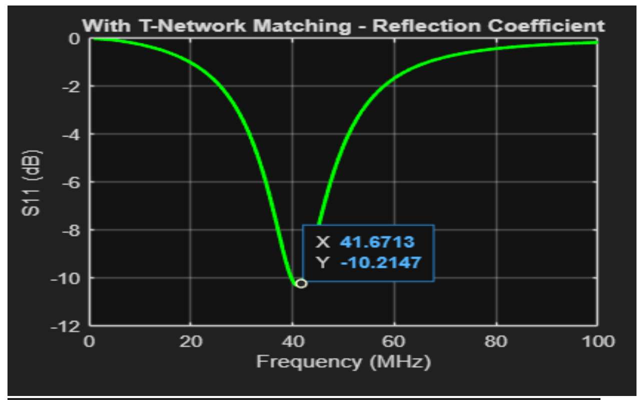
The archived results are:

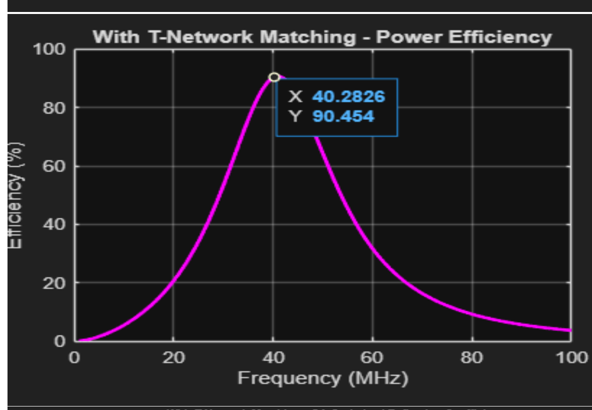
Without T networking

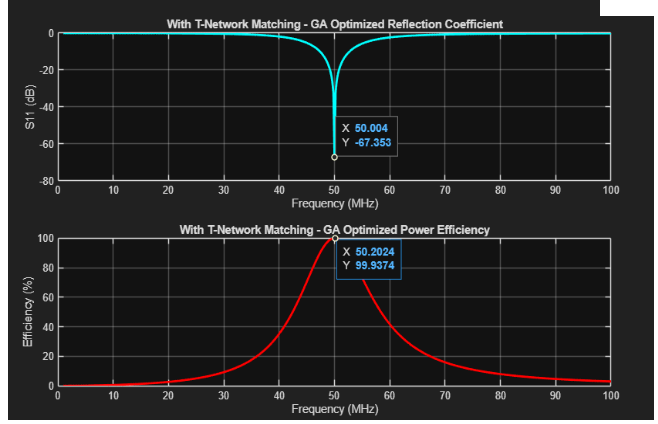




With T networking







Comparative results

	Without T network	With T network
Resonant Freq (MHz)	—	41.08
Minimum S11 (dB)	-3.18	-10.31
Efficiency (%)	51.90	90.69

Optimized sweep result

Resonant Freq (MHz)	*50.00*
Minimum S11 (dB)	*-67.35*
Efficiency (%)	*100*

The results clearly represent that using a T-networking impedance matching circuit greatly improves the performance of RF coil when connected to biological loads. The better values of return loss (S11) and power transfer efficiency prove that the designed circuit reduces signal reflection and allows more power to pass smoothly from the source to the load

T-Network Impedance Matching Circuit was modeled using three stages: Unmatched, Manually Tuned, and Optimization Tuned. The Unmatched and Manually Tuned stages are used as a reference point while the Optimization Tuned stage uses the tuned component values from the optimization algorithm to calculate the best that can be achieved in terms of system performance

VI . Conclusion

The paper reveals that the impedance of the T-network matching circuit is useful in terms of equivalency of the impedance between the RF coils and the biological load. This is matched, which decreases signal loss and enhances efficiency of power transfer. The findings demonstrate that the developed design can be utilized in the biomedical sensors to provide an improved transmission of sensors and credible performance.
Optimization concept (in the form of Genetic Algorithm) was used within this study for automatic identification of maximized T-Network component values C1, L, C2. The sophisticated, automated tuning resulted in performance better than design made manually with highest possible efficiency being reached.

VII. References

[1] S.-M. Sohn, L. DelaBarre, A. Gopinath, J. Vaughan and R. W. Brown, Design of an electrically automated RF transceiver head coil in MRI, IEEE Transactions on Biomedical Circuits and Systems, vol. 8, no. 6, pp. 788 796, Dec. 2014.

[2] S. K. Kandala, K. P. V. S. S. R. Meka and M. R. Islam, "Design of standalone wireless impedance matching (SWIM) system of RF coils in MRI," Scientific Reports, vol. 12, 2022

. [3] K. Byron, A. J. Miller, M. J. Ehrlich and S. M. Reznik, "An RF MEMS controlled wireless power and matching system that is also compatible with MRI systems is obtained using an MRI-compatible entrepreneurial system, 2019, IEEE Transactions on Biomedical Engineering.

[4] F. D. Hockett et al., Simultaneous dual-frequency 1H and 19F open coil design and return-loss measurements with phantom loads, in MAGMA (Magnetic Resonance Materials in Physics, Biology and Medicine), 2010.

[5] W. Wang, Three-element matching networks to decouple receive-only MRI coils. Technical Report Thesis, DTU, 2021.

[6] A. Mohan, Implantable antennas to use in biomedicines: a review, Bioelectromagnetics. Open Access review (PMC), 2024.

[7] A. Fereshtian, A. M. Sodagar and M. Mousavi, "Impedance matching and efficiency enhancement of a dual-band wireless power transfer system with variable inductance, Elsevier International Journal, 2020.

[8] V. Yashchenko, V. Turgaliev and D. Kozlov, Adaptive impedance matching network of wireless power transfer system with offcenter receiver PIERS 2017, May 2017.

[9] K. A. Thackston, Optimization of wireless power networks in biomedical implants M.S. thesis, Purdue University, 2018.

[10] S. Miao, Z. Guo and Y. Zhang, With closed-loop control, an adaptive impedance matching network with IWPT, Sensors, vol. 17, no. 8, 2017.

[11] R. K. Ahire and D. P. Gaikwad, "Compensation topologies of inductive wireless power transmission to implantable biomedical devices a review," Elsevier, Electronics, 2022.

[12] X. Cao et al., "Direct matching methods of coils and preamplifiers in MRI," Journal, Review, 2018.

[13] S. M. Reznik and J. T. Vaughan, Remote and automated matching networks S. M. Reznik and J. T. Vaughan, US Patent IEEE Proceedings discussion, 2014, 2016.

[14] L. Regnacq " A portable and multimodal system of biological impedance spectroscopy measuring complex loads and matching 2023.

[15] K. Payne "Dual-tuned coaxial transmission line RF coils: design, matching and applications: Magnetic Resonance Materials in Physics, Biology and Medicine 2023.

[16] S. D. Pozar, Microwave Engineering, 4th ed. Wiley ch. on matching networks (typical text coverage of L,T,3 networks).

[17] D. M. Pozar, R. C. Johnson texts on papers regarding three element and multi element matching networks of narrowband, wideband matching (IEEE Trans. MTT and IEEE Proceedings).

[18] Pescovitz, A. G. and Brown, B. H., Techniques of measurement of coil load and matching by return loss (S11), Measurement, Instrumentation proceedings 2015.

[19] S. Zhang, J. Liang and H. Gu, T-type dynamic impedance matching circuit design of piezoelectric transducers (principles that

apply to RF matching of complex loads), SPIE Proc 2020.
[20] J. Rieke and K. Butts Pauly, MR physics: coil tuning and matching practical overview, educational proceedings (ISMRM/teaching) 2013.
[21] H. L. Thakur has Designed strategies of robust coil matching under the variable of biological loading, IEEE ISAP 2016, 2019
[22] M. A. Stuchly and S. S. Stuchly, On the dielectric properties of biological tissues and their influence on antenna, coil matching, classic article, IEEE Transactions on Biomedical Engineering.
[23] T. D. O'Connor et al, Preamplifier decoupling and three-element matching of MRI receive arrays, ISMRM, IEEE MR hardware sessions, 2019.
[24] S. L. Wang, J. Lin, H. Zhang, H. Shi, H. Cheng, H. Kuang, H. Chen, and H. Tao, Optimization and Automatic Tuning of RF Coil Matching Networks to Wearable Biomedical Sensors, IEEE Sensors Journal Conference, 2021.
[25] K. R. Foster and S. C. Gabriel, Electrical properties of tissue: implications of RF coil loading and matching, Review article, Biological Effects of EM fields.